TO 33kV SWITCHBOARD F00-SW-010 PANEL B23
REFER 2000-F00-SLD-EL-10006

INTERTRIPS TO SIGNALS

F23-NR-016
FINAL TAILS TRANSFORMER
F23-TR-016 NEUTRAL EARTHING CUBICLE

F23-TR-016
33 /11 kV
5000 kVA
Dyn11
z=7%
RNAN

254Ω +/-5%
30/0.05A 25A FOR 10s
SENSING
ER-15kV

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

F23-SR-001
FINAL TAILS PUMPS 11kV & 415V SWITCHROOM

E-SHUT F23SR001-LCS
NOTE 5

F23-SW-016
FINAL TAILS PUMPS 11kV SWITCHBOARD

NOTE 4

TEST
300/1A 2.5VA Cl 0.2 — PQM SEL735
PR SE-330AU
EARTH FAULT INDICATION
TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)
UPSTREAM TRIPS
300/1A 2.5VA 5P20
CLOSE
TRIP
630A
NOTE 2
PR SEL 751
50 51 50G 51G AFD 50N 51N 49 63 71
NOTE 3

2000-F23-SCM-EL-10101
NOTE 2
A01 A02 A03 A04 A05 A06
2000-F23-SCM-EL-10107
11kV 1250A 3-E 25kA FOR 3s
110V AC SUPPLY

2A 630A NOTE 2

VT
11kV/√3 0.11kV/√3 0.11kV/√3
15VA CLASS 0.2
20VA CLASS 3P

160E
CLOSE
50/1A 2.5VA 5P20
50/1A 1.5VA 10P10
TEST
VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT

F23PP466-VS
FINAL TAILS PUMP 1 VSD 11/3.3kV 800kW
F23PP467-VS
FINAL TAILS PUMP 2 VSD 11/3.3kV 800kW
F23PP468-VS
FINAL TAILS PUMP 3 VSD 11/3.3kV 800kW
F23PP469-VS
FINAL TAILS PUMP 4 VSD 11/3.3kV 800kW

M 3~ 800kW RTD

F23-PP-466
FINAL TAILS PUMP 1
2000-F23-SCM-EL-10103

F23-PP-467
FINAL TAILS PUMP 2
2000-F23-SCM-EL-10105

F23-PP-468
FINAL TAILS PUMP 3
2000-F23-SCM-EL-10109

F23-PP-469
FINAL TAILS PUMP 4
2000-F23-SCM-EL-10111

CONT'D ON SHEET 2 DRG 2000-F23-SLD-EL-10002

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND UPSTREAM 33kV FEEDER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.
7. UPSTREAM 11kV FEEDER SHALL BE TRIPPED ON THE EARTH & ARC FAULT DETECTION OF THE DOWNSTREAM INCOMER ACB.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
27	UNDER VOLTAGE
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
49	TRANSFORMER TEMPERATURE
63	TRANSFORMER PRESSURE / BUCCHOLZ
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

2000-F23-SLD-EL-10001_Rev1

Single line diagram sheet 1